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The effect of damage on the growth, reproduction and storage of lipids in the scleractinian coral *Pocillopora damicornis* (Linnaeus)

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Abstract

A population of the scleractinian coral *Pocillopora damicornis* (Linnaeus) was examined at Rottnest Island, off the Western Australian coast. This is a temperate zone and an area under considerable tourist pressure. Experiments were conducted at Little Salmon Bay and Mary Cove to investigate the effect of mechanical damage on the growth, reproduction and storage of lipid to see whether patterns of energy allocation were affected. These parameters were measured simultaneously 14 times over 13 months.

Damage had an adverse effect on the performance of the corals. Dislodged corals had increased mortality, decreased growth, decreased reproduction but no change in lipids. Fragmented corals had no significant change in mortality, lipids or growth rate but had increased reproduction. The overall allocation of energy to growth, reproduction and lipid storage of dislodged corals appeared to decrease whereas the overall output of the fragmented corals has increased.

Keywords: Coral damage; Growth lipids; *Pocillopora* reproduction

1. Introduction

Pocillopora damicornis (Linnaeus) forms a small coral reef and occurs on many limestone reefs on Rottnest Island, off the Western Australian coast. This area is close to the southern-most limit for this coral, where natural stress is generally greater (temperature fluctuations, storms etc.). Thus *P. damicornis* on Rottnest Island may be more susceptible to damage caused by human impacts on the island such as snorkelling and boating.

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Growth has been used many times as a measure of the performance of corals (Buddemeier & Kinzie III, 1976). Growth patterns vary a great deal within species according to a host of environmental factors. Daily growth ridges in the epithecae (Barnes, 1972), and high and low density growth bands develop according to the favourability of conditions. Thus, the linear extension of the skeleton can be used as a measure of growth for the coral head. At times there is not a simple relationship between size and age for corals as partial mortality, colony fission and colony fusion may distort measures of growth (Hughes & Jackson, 1980). In this study, care was taken to not sample damaged branches.

Interspecific growth rates may vary a great deal, according to a host of environmental factors, and have been used to indicate the health and status of corals (Buddemeier & Kinzie III, 1976; Brown and Scoffin, 1986). The linear extension rate of the coral skeleton may be measured, using the Alizarin staining technique (Lamberts, 1974), to assess the response of coral to different environmental conditions (Gladfelter et al., 1978) and pollutants (Dodge et al., 1984).

Corals contain relatively large quantities of lipid in their tissues (30–40%) (Patton et al., 1977; Stimson, 1987). The lipids may be either structural or storage (Patton et al., 1977). They are distributed throughout coral tissues and associated with reproductive cells and eggs (Stimson, 1987). Small amounts are also present in mucus produced by some corals (Meikle et al., 1988). Changes in the amount of lipid present in coral tissues may be an indicator of stress. The quantity of lipid decreased in corals during the El Nino warming event of 1983 (Glynn et al., 1985) and may decrease with decreasing irradiance (Stimson, 1987).

The sexual reproductive status of corals can be determined histologically (Kojis & Quinn, 1981a,b) or by the collection of released gametes and planulae (Stimson, 1978; Harriot, 1983; Richmond & Hunter, 1990; Wyers et al., 1991). Gametogenesis may be affected by stress such as prolonged bleaching (Szmant & Gassman, 1990) and sexual processes may be halted or reduced (Brown & Howard, 1985). Decreased reproductive activity may include failure to produce gametes or planulae, diminished gamete and planulae production, and a decreased amount of lipid per planula.

This study aimed to investigate allocation of energy in corals at Rottnest Island. Corals can use energy for three main functions: in growth, reproduction and storage of lipid (Sibly & Calow, 1986). Allocation of input to these three functions was measured simultaneously in a population of corals over a period of 13 months. Growth was measured as linear extension, storage of lipid was estimated gravimetrically as the percentage of tissue made up of lipid and reproductive activity was investigated by measuring ova production, production of planulae and the lipid content of those planulae. Stages of gametogenesis were also monitored.

To elucidate the effects of stress on allocation of coral energy, experiments using mechanical damage were conducted at two sites. Mechanical damage is a prevalent stress to corals at Rottnest Island and this naturally occurring damage was previously monitored to determine the most common types of damage (Ward, 1991). The most common types of damage were fragmentation and dislodgement, so these were used as treatments for the damage experiments and the growth, reproduction and storage of lipid was measured for these corals. This allowed comparison between the patterns

of allocation for the damaged and the undamaged corals as well as between individual factors such as growth rates.

2. Methods

2.1. Sites

Rottnest Island lies 20 km off the coast of Western Australia at 32° S. Two sites were sampled: Little Salmon Bay and Mary Cove. These sites are both on the west side of the island and recieve some direct wave action. Corals grow on limestone but there is no coral reef structure at these sites.

2.2. Experimental design

Manipulative experiments were set up at Mary Cove and Little Salmon Bay. All *P. damicornis* colonies chosen for this study appeared healthy, were between 30 and 50 cm in diameter, and were not shaded. Two treatments were used: fragmentation and dislodgement. Fragmentation involved breaking some branches from the colony with a hammer but leaving the colony in place. Approximately 50% of branches were broken off throughout the colony rather than just one section of the colony. The remaining colony was sampled. Dislodged corals were knocked from the reef onto the substratum below which was either sand, rock or seagrass bed. Corals were placed face up at the time of damage and at each sampling time.

An orthogonal design was used for this experiment so all possible combinations of these two treatments were used. At both sites there were fragmented only, dislodged only, fragmented and dislodged and not fragmented, not dislodged corals (Table 1).

2.3. Sampling procedure

Sampling of colonies involved breaking three branches, ≈ 5 cm long from each coral at each date. Branches from the lowest parts of the colonies were avoided. Samples were used to monitor growth, reproduction and lipid storage. The branch tips were used for growth measurement and the rest of each branch for lipids and evaluation of the

Table 1
Number of corals sampled at Little Salmon Bay and Mary Cove during the 13-month study period. Treatments were dislodgement and fragmentation

	Little Salmon Bay	Mary Cove
Fragmented dislodged	7	5
Not dislodged	6	4
Not fragmented dislodged	4	6
Not dislodged	10	7

gametogenic cycle. Sampling began in July 1988 and continued until September 1989. During the winter, samples were taken each month and during the summer months they were taken every 2 wk.

2.4. Growth

In order to measure rates of linear extension of branches, colonies of *P. damicornis* at both sites were stained in situ using the vital stain sodium alizarin sulfonate. The dye was incorporated into the skeleton of the coral colouring it purple where carbonate deposition occurred. Between July and November, plastic bags were placed over the corals just after dawn and tied in place with long strips of Dayglo tape in order to isolate them from the surrounding sea water. Alizarin was injected into the plastic bags at a concentration of 15 $mg \cdot l^{-1}$. The bags were left in place until just before sunset nine hours later.

Three branches were taken from the colony and the tips were placed into fresh water. After 2–3 days the branch tips were washed with a jet of tap water until all the tissue had been removed. They were then cut on a trim saw with a diamond blade so that there was a flat face on which to measure the linear extension from the stain mark. Once the coral had dried, the growth was measured using callipers (to ± 0.1 mm) under a magnifying lamp.

2.5. Lipids

The remaining branch sections from the above growth samples were placed in 10% formaldehyde in seawater for 24 h. The samples were then immersed in Bouin's fixative for 2 wk to decalcify the skeleton. The tissue samples were placed in embedding cassettes and dried in an oven for 24 h at 55 °C. The samples were then weighed (to ± 0.0001 g) and immersed on a 2:1 chloroform-methanol solution for 24 h to dissolve the lipids (Barnes & Blackstock, 1973; Stimson, 1987). The tissue samples were then redried for 3 h and reweighed. The total lipid content was calculated by subtraction and expressed as a percentage of the total tissue biomass.

2.6. Larval capture

To estimate the number of larvae produced, portions of colonies were taken from each coral 4 days before the new moon in March and transported to the laboratory. Colonies were placed in fresh sea water and larvae were collected, counted and stored in a solution of 10% formaldehyde in sea water.

The number of planulae shed per piece of coral was standardised by calculating the relative surface area of each branch. Branches were weighed, placed in a Miles Tissue Tek consul of paraffin wax at 60 °C for 5 s and then reweighed when cooled. The weight of wax was then calculated and the number of planulae shed was divided by the weight of wax.

Captured larvae were analysed for lipid content following the methods used by Barnes & Blackstock (1973) and Stimson (1987). Larvae from individual colonies were

placed in small vials and dried in an oven at 60 °C for 24 h. They were then covered with a 2:1 solution of chloroform methanol for 24 h to extract the lipid before being dried again for 3 h and reweighed.

2.7. Gametogenesis

The number and position of samples taken for gametogenesis was the same as that of the lipid samples except that extra samples were taken 4 days before and 1 day after the new moon in March. The samples were prepared for histology by placing them in a solution of 10% formaldehyde in sea water for 24 h before being decalcified in Bouins solution for ≈2 wk. Tissue was then peeled off and placed into embedding cassettes. Samples were processed through a graded series of alcohols and then cleared in 100% chloroform. They were then impregnated in paraplast wax and embedded in paraffin wax. Sections of 6 μm were cut and these were stained with Gills Haematoxylin and Eosin.

On each slide all polyps (50–100) were scored for their stage of gametogenesis and their planulae were counted. Stages were identified basically following those of Stoddart (1984a,b). There were four stages of spermatogenesis and two stages of oogenesis.

Reproduction includes both production of planulae and gametes, so a measure of reproductive output which included both components was devised. An index of reproductive potential (IRP) was calculated from the percentage of polyps with ova and planulae for each colony at each date. The largest sum of the percentage of polyps containing the two oogenic stages and/or planulae at each time point of each coral was termed the IRP.

2.8. Statistical analysis

Comparison of growth, IRP and the percentage of lipid in tissue at different sites and with different damage treatments was made using analysis of variance. Homogeneity of variances was tested using Cochran's test and, where necessary, appropriate transformations were performed. The damage experiments were treated as an orthogonal analysis of variance and, as site was included, this was a three-factor design.

3. Results

3.1. Mortality

The most dramatic effect of damaging the corals was on their survival. There was a very low mortality of undamaged colonies at both sites with only one of the 14 undamaged colonies dying. There was a higher mortality amongst the dislodged and fragmented colonies with half of these corals dead by the end of the study period (Table 2). Significantly more dislodged corals died than not dislodged corals and, although more fragmented corals died than not fragmented corals, this difference was not significant.

Table 2
The number of corals* from Little Salmon Bay and Mary Cove which died during the study period

	Fragmented		Not fragmented		Dislodged totals	
	Dead	Alive	Dead	Alive	Dead	Alive
Dislodged	8	5	12	8	20	13
Not dislodged	6	5	1	13	7	18
Fragmented totals	14	10	13	21		
Fragmentation	χ^2 (df = 1) = 2.24 $p > 0.05$		Dislodgement		χ^2 (df = 1) = 5.97 $p < 0.05$	

* These were analysed using contingency tests for independence. Dislodged and undislodged corals were compared for mortality in one test and fragmented and not fragmented corals were compared for mortality in the next test.

3.2. Growth

Dislodged corals grew slower (between December 1988 and June 1989) than those which were not dislodged (Table 3) (Fig. 1) but there was no significant effect of fragmentation on the amount of cumulative growth. Corals at Little Salmon Bay grew faster than those at Mary Cove (Table 3).

3.3. Lipids

There was no clear pattern of the effect of dislodgement, fragmentation or site on the quantity of lipid in tissue of experimental corals at Little Salmon Bay and Mary Cove on the 12 dates the analyses of lipids were conducted. In November the dislodged corals had less lipid in the tissues than the undislodged colonies (Fig. 2) and on the 7th February colonies from Mary Cove had a higher quantity of lipid than those from Little Salmon Bay.

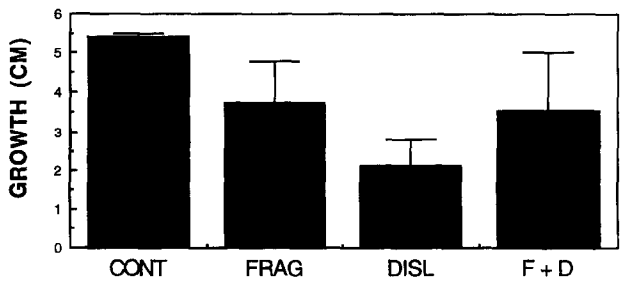

Fig. 1. Growth, as linear extension, of colonies of *P. damicornis* at Rottnest Island. Treatments are cont = controls, frag = fragmentation, disl = dislodged, f + d = fragmentation and dislodgement. Error bars are standard errors.

Table 3
Three-factor orthogonal analysis of variance information for growth (as cumulative linear extension) of corals from Little Salmon Bay and Mary Cove between 17th December and 10th June, 1989

Factor	Mean	SE	N	F	p
Dislodged					
Dislodged	3.5	1.06	12	5.993	0.0237
Not dislodged	5.5	0.66	16		
Fragmented					
Fragmented	4.7	1.17	11	1.325	0.2634
Not fragmented	4.6	0.67	17		
Site					
Little Salmon Bay	5.3	0.84	15	5.085	0.0355
Mary Cove	3.9	0.86	13		
	Interactions		F	p	
	Fragmented–site		0.088	0.7698	
	Dislodged–site		3.285	0.0850	
	Fragmented–dislodged		0.460	0.5056	
	Dislodged–fragmented–site		0.120	0.7302	

Treatments were dislodgement, fragmentation and site. Varances were homogeneous ($C = 0.3103$). N = number of colonies.

3.4. Larval capture

There were numerically fewer larvae captured in dislodged than not dislodged corals, more in fragmented than not fragmented corals and fewer at Mary Cove than at Little Salmon Bay but none of these differences were statistically significant (Fig. 3).

The lipid content of the larvae was most frequently 50–60%. Some corals produced very few larvae and these were difficult to analyse. Although there was less lipid in the larvae captured from corals which were dislodged than those which were not dislodged and more in not fragmented than fragmented corals, the differences were not significant (Fig. 4). The data for larval capture were significantly correlated with the number of planulae from the histological data, although the relationship is not strong ($r = 0.56$, $p < 0.05$, $n = 46$).

The index of reproductive potential (IRP) was significantly larger in fragmented than unfragmented corals. In dislodged corals the IRP was smaller than that of the corals which were not dislodged but not significantly so. There was no significant effect of site (Table 4). These trends are the same as those found for the data on capture of larvae shed in the laboratory (Fig. 4).

Corals with the four damage treatments: fragmentation only, dislodgement only, dislodgement and fragmentation and no damage followed similar gametogenesis patterns. All groups showed the disappearance of late sperm and ova on 7th February and 5th April and all had major planulation in March (Ward, 1992). All four groups planulated in February as well as March although the February event was considerably smaller.

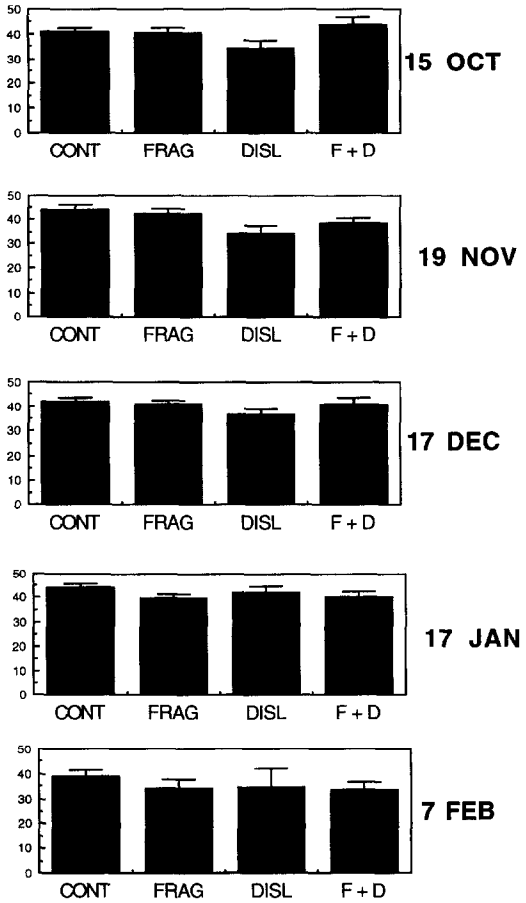

Fig. 2

4. Discussion

Damaging the corals clearly affected their performance. Firstly, dislodgement caused high mortality with the majority of dislodged corals dying by the end of the study period. Dislodged corals also had slower growth. A dislodged coral which lays in sand or on sea grass may be exposed to increased sedimentation and less light. Without its usual attachment, the coral is more susceptible to damage through water movement as it may be carried across the substratum, causing increased abrasion and sections of the colony to face into the sand. Colonies may be battered against rocks or thrown into nearby sea grass beds. Dislodged corals in this study sometimes died very quickly following a storm, perhaps as a result of shut-down reaction occurring (Antonius, 1981, 1988).

Mortality was high in the fragmented corals. Fragmentation had no obvious effect on the cumulative growth of colonies, which is contrary to findings of Simpson (1987) who reported that growth of colonies damaged by a tropical cyclone was slower after

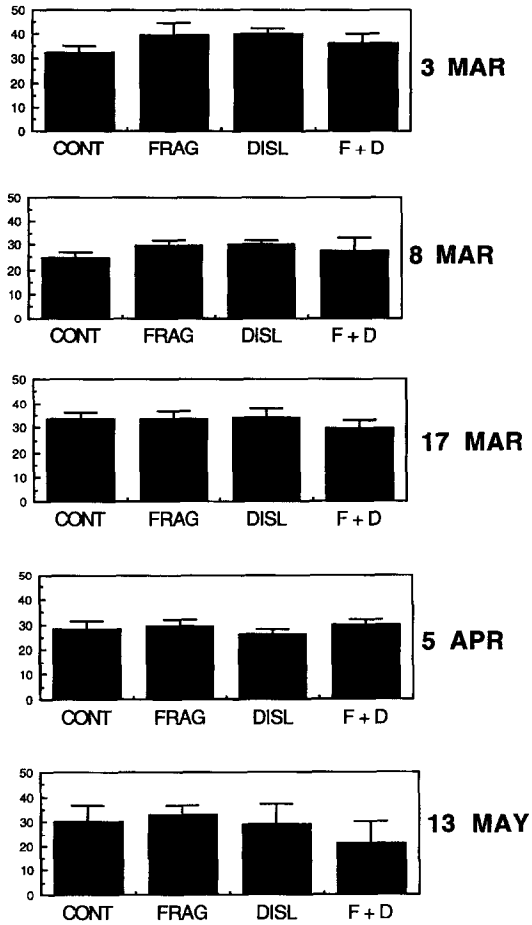

Fig. 2. The percentage of tissue of *P. damicornis* colonies made up by lipid on sampling dates from 15th October 1988 to 5th April, 1989. Treatments are cont = controls, frag = fragmentation, disl = dislodged, f + d = fragmentation and dislodgement. Error bars are standard errors.

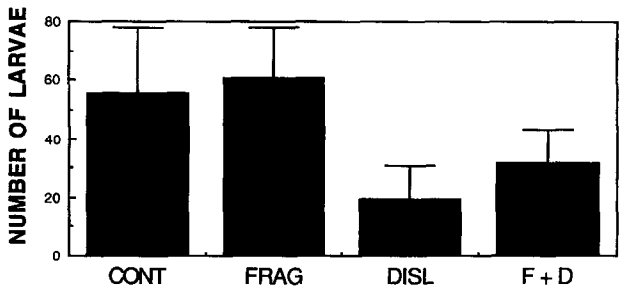

Fig. 3. The standardised number of larvae captured at the new moon in March, 1989 from colonies of *P. damicornis* at Rottnest Island. Treatments are cont = controls, frag = fragmentation, disl = dislodged, f + d = fragmentation and dislodgement. Error bars are standard errors.

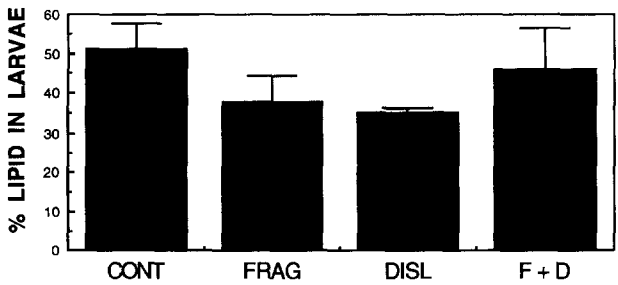

Fig. 4. The percentage dry tissue weight made up by lipid of larvae captured at the new moon in March, 1989 from colonies of *P. damicornis*. Treatments are cont = controls, frag = fragmentation, disl = dislodged, f + d = fragmentation and dislodgement. Error bars are standard errors.

Table 4
Analysis of variance comparing index of reproductive potential, for fragmented, dislodged and control corals from Little Salmon Bay (LSB) and Mary Cove (MC)

Factor		Mean	SE	*N*	*F*	*p*
Dislodged	yes	42.98	4.98	17	3.77	0.0623
	no	57.15	4.43	19		
Fragmented	yes	59.58	4.55	15	6.047	0.0204
	no	43.94	4.57	21		
Site	LSB	45.92	4.94	19	2.381	0.1341
	MC	55.54	4.73	17		

Interactions	*F*	*p*
Fragmented–site	1.692	0.2039
Dislodged–site	0.003	0.9573
Fragmented–dislodged	0.013	0.9084
Dislodged–fragmented–site	0.607	0.4423

Variances were homogeneous ($C = 0.2211$).

the cyclone than it had been before the cyclone. Initially the fragments from fragmented colonies in this study were monitored and showed a trend of low growth but these either died or disappeared rapidly. Bak & Criens (1981) conducted a fragmentation experiment and found an initial decrease in living coral cover followed by overgrowth by living coral. Towards the end of the experiment, growth was less than the control quadrant but mortality was very low until an outbreak of a disease killed all the fragments. During this study, there was increased mortality of fragmented corals following one large storm near the end of 1988. Before this storm, mortality of fragmented corals was very low. Fragmentation may make colonies more susceptible to sudden events such as storms and disease outbreaks.

With respect to growth and mortality, dislodgement appears to be a more severe type of damage than fragmentation, as it caused a decrease in cumulative growth as well as an increase in mortality.

Corals at Rottnest Island had a high lipid content. The 29–46% of tissue weight made up by lipid recorded here is slightly higher than past studies with Patton et al. (1977) finding 34% for *P. capitata* and Stimson (1987) finding 30–40% for *P. damicornis*. It is possible that the high percentage of lipid found in this study may be a characteristic of corals in cooler water. However, there are no suitable studies with which to compare these results.

There were fluctuations in the amount of lipid present in the tissue of the sampled corals throughout the year. Corals had more lipids in their tissues between August and February than between March and the following August. On a finer scale the quantity of lipid decreased at the February, March and April new moons and had increased by the sampling period 2 wk after these new moons. These reductions coincide with the loss of planulae and gametes over these periods. Damage did not obviously affect the quantity of lipids in the coral tissue as there was no difference between treatments in the quantity of lipid present on any date, or either pre- or post-planulation.

The reproductive activity of *P. damicornis* at Rottnest Island was affected by both dislodgement and fragmentation. The IRP was lower for dislodged corals due to reduced numbers of both ova and planulae. Dislodged corals are frequently exposed to greater sediment stress, due to resuspension, because they are on the substrate surrounded by sand. This may cause a decrease in light available to the corals which is often compounded by a slight decrease in light due to increased depth as the corals usually fall from the reef downwards.

An increase in the IRP resulted from fragmenting the colonies. Fragmention during this study involved removing branches throughout the colony so this may have caused an increase in light available to many of the remaining branches. Kojis & Quinn (1984) found that fecundity in *Acropora palifera* varied proportionately with high levels and correlated with water clarity and sedimentation. The most fecund section of the branch is the mid section below the growing tips and in an intact colony these sections are often in relative shade. Rinkevich (1989) used labelled carbon to conclude that the energy requirements for planulae production in hermatypic corals are supported by translocation of products from algal photosynthesis. Fragmentation may have increased the light available for the fecund areas and this may have caused the increase in reproductive potential.

Pocillopora damicornis at Rottnest Island follows the pattern of this and other coral species at other temperate zone and subtropical reefs in that growth is fastest during the warmest months, in this case February, March and April (Stoddart, 1984a; Ward 1991). This is in contrast to tropical reefs where growth is usually fastest during the winter months (Highsmith, 1979). The period of increased growth at Rottnest Island is short with only 3 months growth which is markedly faster than any other time.

No comparable data on recovery rates for fragmentation and dislodgement in tropical areas, using the parameters measured in this study could be found, but it seems likely that recovery would be faster in the tropics where growth is faster. Kay & Liddle (1989) studied the effects of trampling by tourists on areas of coral reef off Heron Island on the Great Barrier Reef and this involved considerable fragmentation of colonies. They found different levels of destruction on different sections of the reef, but did not look closely at recovery rates so a comparison is difficult. In the relatively warm water of

Costa Rica, Guzman & Cortes (1989) found that *Porites lobata* used fragmentation as a dispersal mechanism and exhibited rapid injury recovery and high resistance to environmental stress.

The results of damage experiments involving dislodgement and fragmentation generally reflected the results of the monitoring of 'naturally' occurring damage on the island (Ward, 1991). Generally, corals which were found dislodged died or did not appear to grow noticeably. Often, they would have abraded sections where the coral had faced the substratum and this area would become overgrown with algae. In contrast, colonies which were found fragmented either died or appeared to have a full recovery.

Gray (1989) suggested that the application of a stressor may cause an alarm reaction followed by compensation and a return to normal and that, if the application continues, the organism may die. He hypothesizes that initially an organism will divert energy from growth into reproduction but on further application of the stressor the fecundity will decrease. Thompson (1983, 1984) found that sea urchins on a depleted food ration decreased their gametic output but, when considered with their accompanying decrease in growth, this represented an increase in reproductive effort. As the mortality figures suggest, the dislodged corals were dying or at least not performing well. These results suggest that they may have shifted allocation away from growth to reproduction. Although the IRP was reduced in comparison with corals which were not dislodged, this may not reflect the reproductive effort of the corals as the proportion of total energy used for reproduction may have increased.

The IRP of the fragmented corals was increased due to more brooded larvae rather than large numbers of ova. The increase in reproductive potential does not appear to be a trade-off reaction to stress as neither the growth nor the content of lipid simultaneously decreased.

As discussed in Ward (1991), there are two forms of allocation operating in corals at Rottnest Island. Corals, which produced brooded larvae produced ova, showed a decrease in growth, an increase in reproductive lipid and a decrease in storage lipid. Corals which did not produce brooded larvae did not produce ova, showed more rapid growth, had less reproductive lipid and more storage lipid. Now the allocation of damaged corals can be compared with the "normal pattern".

Although the seasonal allocation patterns were basically the same for damaged corals, some differences were apparent. For dislodged corals almost all produced brooded larvae and there was, overall, a reduced output. Growth was decreased and there was a reduction in reproductive potential and no change in lipids. In contrast, the fragmented colonies appeared to increase their output. There was no significant change in cumulative growth but reproductive potential was increased and there was no change in lipids.

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